

Hu-CEMNIST: A Benchmark Dataset of Human-Generated Counterfactual Explanations for MNIST

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Abstract

Counterfactual explanations are commonly evaluated using criteria such as sparsity, proximity, and plausibility, while less is known about the counterfactual changes people themselves consider meaningful. We introduce Hu-CEMNIST, a benchmark dataset of 4,659 human-generated counterfactuals covering all 90 non-identical MNIST digit transformations. Participants minimally edited a representative source digit to make it recognizable as a specified target. The resulting images were quality-controlled by a minimum-change criterion and retained only when three independent human raters agreed on the target digit. We compare these edits with counterfactuals generated by DiCE, prototype-guided counterfactuals, and normalizing-flow-based counterfactuals. Our results indicate systematic differences between human and algorithmic counterfactuals: some algorithmic methods produce images closer to the MNIST training distribution, whereas others produce adversarially effective perturbations, whereas human edits concentrate on visually meaningful digit strokes. Hu-CEMNIST provides a benchmark for evaluating whether counterfactual generators align not only with classifier decisions, but also with human intuitions about meaningful visual change, thus supporting purpose-dependent evaluation by complementing model-centered criteria.

1 Introduction

Counterfactual explanations (CFE) [Wachter *et al.*, 2017] have become a central paradigm in explainable artificial intelligence (XAI) [Guidotti, 2024; Stepin *et al.*, 2021; Wang *et al.*, 2021]. Rather than describing a model globally, they explain a particular prediction by showing how the input would need to change for a different output to occur [Mothilal *et al.*, 2020]. In image classification, a CFE may answer a question such as: *Which changes to an input image would make a classifier assign it to another target class?* [Zhao *et al.*, 2021]. This form of explanation is appealing because it is contrastive, local, and easy to communicate [Byrne, 2019],

although empirical support for its benefits on human predictive accuracy remains mixed [Warren *et al.*, 2024].

Most counterfactual generation methods operationalize explanation quality through computational criteria such as proximity to the original input, sparsity of the edit, validity with respect to the classifier, or plausibility under the data distribution [Verma *et al.*, 2024; Artelt and Hammer, 2020; Wachter *et al.*, 2017]. These criteria are useful, but they do not necessarily capture how humans themselves construct counterfactuals in visual domains. In particular, prior work comparing human- and machine-generated CFEs for images suggests that people do not simply make the smallest possible pixel-level changes. Instead, they tend to make larger but semantically meaningful edits that move an image toward a recognizable prototype of the target class [Delaney *et al.*, 2023]. This is consistent with a broader human-centered evaluation perspective in XAI, where human-generated explanations and annotations are treated as empirical reference standards for assessing explanation quality [Mohseni *et al.*, 2021; Kazmierczak *et al.*, 2026; Zhang *et al.*, 2025].

Human-centered evidence remains sparse, underscoring the gap between theory and empirical validation [Keane *et al.*, 2021]. Yet, previous work has taken an important first step by collecting human CFEs for misclassified images, albeit at a relatively small scale [Delaney *et al.*, 2023]. Building on this foundation, we extend the human-centered counterfactual benchmark setting to a larger, systematically controlled set of source–target digit transformations.

Benchmark-design work in the area of remote-sensing emphasizes the importance of scale, coverage, and annotation quality control [Long *et al.*, 2021; Shi *et al.*, 2023], and similar conclusions arise in text annotation, where large-scale analyses of dataset publications reveal frequent annotation errors without robust quality-control procedures [Klie *et al.*, 2024]. We adopt the same rationale here: systematic CFE benchmarking requires many human-generated counterfactuals across digit transformations, together with checks ensuring that images are both changed and recognizable. MNIST is a useful starting point for such a benchmark, as it is simple, widely used [Looveren and Klaise, 2021; Kenny *et al.*, 2021], visually interpretable, and has a clear class structure, while still allowing many different counterfactual transformations between digits.

Our contributions are threefold: First, we introduce Hu-

CEMNIST, a benchmark dataset of human-generated counterfactual transformations covering all 90 non-identical ordered digit pairs. Second, we describe a scalable acquisition and quality-control pipeline for collecting such visual counterfactuals from human participants. Third, we compare the resulting human-generated counterfactuals with counterfactuals generated by established algorithmic methods, using metrics that capture changed pixels, similarity to MNIST training examples, and similarity to human edits.

The results show that current algorithmic counterfactual generators can differ substantially from human-generated edits, adding to broader evidence of persistent gaps between explanation plausibility, user satisfaction, and actual human understanding [Brughmans *et al.*, 2023]. Such differences are not inherently problematic. Counterfactuals designed to reveal a classifier’s local decision boundary may appropriately diverge from human transformations, whereas explanations intended to communicate semantically meaningful changes may benefit from closer agreement. Hu-CEMNIST therefore supports a purpose-dependent evaluation, with human alignment complementing model-centered criteria such as validity and proximity.

2 Foundations

A CFE [Wachter *et al.*, 2017] describes how an input instance would need to change in order to obtain a different prediction, thereby answering the contrastive question: why this class rather than another [Laugel *et al.*, 2023]? In image classification, let x denote an input image and let $h(x) = y$ be the prediction of a classifier. A counterfactual image x_{cf} for target class y_{cf} should satisfy $h(x_{cf}) = y_{cf} \neq y$, while remaining as similar as possible to the original image x and being a plausible instance in the data domain. This is commonly formalized as a multi-objective optimization problem that balances prediction change against constraints such as plausibility and the magnitude of the required edit [Del Ser *et al.*, 2024]. In the context of MNIST, x is a handwritten digit image, y is its label, and y_{cf} is the desired target digit.

CFEs are usually evaluated according to validity, proximity, sparsity, and plausibility [Verma *et al.*, 2024; Guidotti, 2024]. Validity requires that the counterfactual actually changes the model prediction to the target class; it is commonly treated as a core evaluation criterion and can vary substantially across methods [Forel *et al.*, 2024]. Proximity requires that the counterfactual remains close to the original instance, often measured by a p-norm [Guidotti, 2024], or pixel-change distance. Sparsity requires that only a small number of features or pixels are changed [Sharma *et al.*, 2024]. Plausibility requires that the counterfactual resemble a realistic data point, i.e., being distributionally coherent [Connell and Keane, 2019], rather than an arbitrary perturbation. In image domains, plausibility is especially important because small pixel-level changes may affect a classifier while remaining imperceptible or meaningless to humans [Zhao *et al.*, 2022; Smyth and Keane, 2022; Artelt and Hammer, 2020].

These criteria are not always aligned, and specific trade-offs exist [Brughmans *et al.*, 2024; Dastile and Celik, 2024].

For image data, a counterfactual can be sparse and valid but visually implausible. Conversely, a visually meaningful human edit may alter more pixels than a sparse optimization method would prefer. However, pixel-wise distances are known to be limited proxies for perceptual similarity and often fail to capture semantic image structure [Zhang *et al.*, 2018; Fu *et al.*, 2023]. This trade-off motivates Hu-CEMNIST: if humans transform digits through semantic stroke edits, then human alignment cannot be evaluated by pixel distance alone.

Prior work has emphasized this distinction between machine- and human-generated counterfactuals. In image-based user studies, humans have been found to produce counterfactuals that are not minimal pixel edits, but rather larger edits that make the image more representative of the target class [Delaney *et al.*, 2023]. This points towards different explanation goals: algorithmic methods often optimize for class-discriminating changes near a decision boundary [Mothilal *et al.*, 2020], whereas human explanations may aim to communicate what makes an instance recognizable as belonging to the target class [Qi *et al.*, 2024]. In fact, recent XAI work increasingly treats human annotations as empirical ground truth for evaluating explanation quality, including human explanation datasets, attention-mask benchmarks, saliency benchmarks, and human-centered XAI evaluation frameworks [Mohseni *et al.*, 2021; Kazmierczak *et al.*, 2026; Zhang *et al.*, 2025]. Thus, instead of treating human plausibility as an abstract concept, we collect human counterfactual edits directly and use them as reference data with explicit quality-control procedures [Klie *et al.*, 2024].

This enables three complementary evaluations. First, human-generated counterfactuals can be analyzed as a dataset in their own right, revealing which image regions people tend to modify for each digit transformation. Second, algorithmically generated counterfactuals can be compared to human-generated ones, both in terms of their final visual appearance and in terms of their change maps. Third, the human-generated counterfactuals can themselves be assessed using conventional CFE criteria, such as proximity, sparsity, and similarity to the data distribution. This reverses the usual comparison and reveals where established computational criteria do or do not reflect human transformation strategies.

3 Benchmark Acquisition

3.1 Selection of MNIST Samples

For each digit in the MNIST dataset (0–9), we compute a pixel-wise average to obtain a mean image that captures the central visual tendency of that digit. To reduce noise and emphasize salient features, we apply a threshold on each mean image, setting all pixel values below 125 to zero. Next, for each digit, we compute the Structural Similarity Index (SSIM) [Wang *et al.*, 2004] between the thresholded mean image and all individual samples corresponding to that digit. The SSIM is a metric used to assess the perceptual similarity between two images. Unlike basic metrics like the $L1$ or $L2$ norms that compare pixel differences directly, SSIM is designed to mimic the human visual perception by considering image luminance, contrast, and structure. The SSIM between

two images x and y is computed as:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (1)$$

Where μ_x and μ_y are the mean values of the images x and y , σ_x^2 and σ_y^2 are the variances of x and y , σ_{xy} is the covariance between x and y , $C_1 = (K_1L)^2$ and $C_2 = (K_2L)^2$ are small constants to stabilize the division with weak denominator values, where L is the dynamic range of the pixel values (typically $L = 255$ for 8-bit images), and $K_1 = 0.01$ and $K_2 = 0.03$ are typical values for the constants. $SSIM$ values range from -1 to 1, where 1 indicates perfect similarity. Higher $SSIM$ values imply greater structural similarity between the two images.

The sample with the highest $SSIM$ score is then selected as the target exemplar, as it best preserves the representative structural features of the digit class.

3.2 Data Acquisition and Quality Control

Data were collected on Prolific using a custom web-based drawing tool that allowed participants to add, erase, and undo pixel-level edits. Each participant completed three digit-transformation tasks. For each task, participants were instructed to transform the source digit into the specified target digit while changing as little as possible and modifying at least 2% of the pixels. The interface displayed the current percentage of changed pixels. The 2% threshold prevented null or near-null submissions while retaining the emphasis on minimal, meaningful edits.

Participants reviewed the task information, gave consent, watched a two-minute instructional video, and completed four comprehension questions, with two attempts allowed per question. Failing twice led to screening out and partial compensation. After three annotation tasks, they completed demographic and feedback questions, including one attention check and three optional open-ended questions on annotation strategy, the 2% minimal-change requirement, and possible improvements to the tool. Recruitment was restricted to desktop users fluent in English and a 100% Prolific approval rate, with balanced male/female sampling. Each participant could enter only once. Each participant received three randomly sampled source–target transformations without replacement. Across acquisition waves, quotas were monitored per transformation, with targeted additional collection for transformations yielding fewer than 50 valid submissions or lower recognition rates.

Further details on the interface, procedure, and post-task survey items are provided in the supplementary material.

We aimed to collect at least 50 valid counterfactuals for each of the 90 non-identical source-target transformations ($9 \times 50 = 4,500$). Collection proceeded in 13 waves, enabling intermediate quality checks and targeted re-acquisition of underrepresented or low-quality transformations. Participants completed the study in a median of 10.5 minutes (range: 9–16) and received GBP 1.50 compensation. The study was approved by the Ethics Committee of Bielefeld University.

All submissions passing the 2% change threshold were rated by three independent annotators. Annotators saw only

Table 1: Demographic composition before and after data cleaning.

	Full sample $N = 2,103$	Final sample $N = 1,961$
Gender		
Male	996 (47.4%)	955 (48.7%)
Female	1034 (49.2%)	961 (49.0%)
Non-binary	13 (0.6%)	13 (0.7%)
Transgender	7 (0.3%)	5 (0.3%)
Other	2 (0.1%)	2 (0.1%)
Prefer not to say	51 (2.4%)	25 (1.3%)
Age		
18–24 years	357 (17.0%)	340 (17.3%)
25–34 years	732 (34.8%)	687 (35.0%)
35–44 years	475 (22.6%)	452 (23.0%)
45–54 years	286 (13.6%)	266 (13.6%)
55–65 years	158 (7.5%)	150 (7.6%)
65 years and older	46 (2.2%)	0 (0.0%)
Prefer not to say	49 (2.3%)	66 (3.4%)

the edited image, without source or target labels, and identified the depicted digit. They could additionally mark responses as uncertain or unrecognizable. We used human recognizability rather than classifier agreement as the inclusion criterion to preserve visually meaningful transformations independently of any specific classifier. Filtering the dataset through the same classifier later used for algorithmic comparisons would risk conflating human recognizability with model-specific validity.

3.3 Results of Acquisition

Of 2,325 participants entering the study, 222 failed comprehension checks. Among the remaining 2103 participants, 4 had no logged data as participant-side connectivity failures prevented data transmission and 11 made insufficient edits in all trials. Thus, 2,088 participants contributed at least one feasible transition, yielding 5,877 images for annotation. Annotators showed similar misclassification rates (R1: 11.74%, R2: 12.22%, R3: 12.45%); with R3 being the most conservative and marking 4.81% of images as unrecognizable (R1: 2.91%, R2: 3.15%). The final Hu-CEMIST dataset contains 4,659 images with perfect agreement across all three annotators, contributed by 1,961 participants (Table 1). For 177 participants, connectivity was disrupted after demographic data transmission, leaving their post-task survey responses unavailable; complete demographic and post-task survey data were recorded for the remaining 1,784 participants.

4 Experiments

For the same digits as used in the user study (see Section 3), we empirically evaluate algorithmically generated counterfactuals of three popular counterfactual generation methods from the literature – i.e., each human trial is complemented by an “algorithmic trial”: DIVERSE Counterfactual Explanations (DiCE) [Mothilal *et al.*, 2020], counterfactuals guided by prototypes (Proto) [Looveren and Klaise, 2021], and counterfactuals based on Normalizing Flows [Wielopolski *et al.*, 2024]. Note that those methods are non-deterministic, and therefore

(can) yield slightly different counterfactuals for exactly the same input – like for the human-generated CFs, we aggregate those non-deterministic CFs by computing an average. To generate algorithmic counterfactuals, we train a Multi-Layer-Perceptron (MLP) with a single hidden layer (of size 128) as a digit classifier – it achieves an accuracy of 97.7% on the test data.

The implementations of the experiments, as well as the data from the user study (Section 3) are available on GitHub¹.

All counterfactuals, whether algorithmically- or human-generated, are evaluated with the following metrics:

- Percentage of pixels being changed:

$$\frac{\sum_{i=0}^{783} \mathbb{1}((\vec{x})_i \neq (\vec{x}_{\text{cf}})_i)}{784} \quad (2)$$

Recall that we have $28 \times 28 = 784$ pixels per digit image.

- SSIM to the closest matching counterfactual digit from the MNIST training data (i.e., training-set similarity), as a proxy for the

$$\max_{\vec{x}_i \in \mathcal{D}_{\text{train}}(y_{\text{cf}})} \text{SSIM}(\vec{x}_i, \vec{x}_{\text{cf}}) \quad (3)$$

where $\mathcal{D}_{\text{train}}(y_{\text{cf}})$ refers to the set of all training samples labeled as y_{cf} .

- Change map, i.e. indices of pixels being changed:

$$\mathcal{M}(\vec{x}, \vec{x}_{\text{cf}}) = \{i \mid (\vec{x})_i \neq (\vec{x}_{\text{cf}})_i \forall i \in \{0, \dots, 783\}\} \quad (4)$$

Most importantly, we also compare them to the human-generated counterfactuals from our introduced Hu-CEMNIST benchmark (Section 3), investigating the difference between algorithmically- and human-generated counterfactuals. For that, we consider the following additional metrics:

- Agreement with human changes, i.e., difference (in percentage) between the change map and the closest matching change map from the human-generated counterfactuals:

$$\max_{\vec{x}'_{\text{cf}} \in \mathcal{D}_{\text{CF}}(y, y_{\text{cf}})} \frac{|\mathcal{M}(\vec{x}, \vec{x}_{\text{cf}}) \cap \mathcal{M}(\vec{x}, \vec{x}'_{\text{cf}})|}{|\mathcal{M}(\vec{x}, \vec{x}_{\text{cf}}) \cup \mathcal{M}(\vec{x}, \vec{x}'_{\text{cf}})|} \quad (5)$$

where $\mathcal{D}_{\text{CF}}(y, y_{\text{cf}})$ refers to the set of human-generated counterfactuals (from the user study) that change the label from y to y_{cf} – note that this always refers to the same original digit though, since there was only one original digit per case in the user study.

- Similarity with the closest human-generated counterfactual, i.e., SSIM to the closest matching human-generated counterfactual from the user study:

$$\max_{\vec{x}'_{\text{cf}} \in \mathcal{D}_{\text{CF}}(y, y_{\text{cf}})} \text{SSIM}(\vec{x}'_{\text{cf}}, \vec{x}_{\text{cf}}) \quad (6)$$

These two human-centered metrics are conceptually related, but capture different aspects of alignment.

To complement descriptive statistics, we conduct a set of inferential and comparative analyses. We treated each source-target digit transformation as a repeated-measures unit and compared counterfactual sources on all four metrics. Because human agreement and similarity are defined relative to human edits, the human-generated condition was structurally unavailable for these metrics; therefore, Friedman tests for these two metrics included only DiCE, Normalizing Flows, and Proto. For training-set similarity and percentage of change, the human-generated condition was retained, yielding four-group repeated-measures comparisons. We used Friedman tests for omnibus effects, Holm-corrected paired Wilcoxon signed-rank tests for post-hoc contrasts, Kendall’s W for omnibus effect sizes, Wilcoxon r for pairwise effects, and Spearman correlations to characterize relationships among metrics.

4.1 Results

The results for the average area of change (i.e., a map showing which pixels have been changed) are shown in Figure 1, and the results of the average similarity (i.e., max SSIM score) of the generated counterfactuals with the MNIST training data are shown in Figure 2. Extended descriptives, statistical tests, and metric visualizations are provided in the supplementary material.

Figure 1 shows *distinct spatial change patterns across counterfactual sources*. Human edits concentrate around digit strokes, whereas DiCE and Proto produce more method-specific localized patterns. The apparently featureless Normalizing Flows panel does not indicate an absence of change: because this method changes nearly every pixel, its change-frequency map is approximately uniform across the image.

Training-set similarity further shows that DiCE and Proto produce results that are most similar to MNIST training samples, whereas human-generated CFs occupy an intermediate position and Normalizing Flows shows the lowest similarity to the target-class training data. Thus, training-set similarity should be interpreted as a related but distinct criterion. Within each algorithmic source, counterfactuals with higher training-set similarity also tended to be more similar to individual human-generated CFs (see Supplementary Table 23 and Supplementary Figure 5). However, at the group level, human-generated CFs did not show the highest training-set similarity; DiCE and Proto were closer to target-class MNIST training samples (see Supplementary Figure 4). Normalizing Flows often produces diffuse changes across all pixels, yielding images that are not typical for the requested target class.

Despite being non-deterministic, all algorithms produce less variable outputs than humans. While the algorithms share a fixed classifier and method-specific objective, participants may rely on heterogeneous internal representations of what constitutes a recognizable target digit. Identifying distinct human strategies requires an explicit clustering analysis.

The statistical analysis confirmed systematic differences between human-generated and algorithmic CFs (Table 2). Human agreement differed strongly across algorithms, with DiCE and Proto outperforming Normalizing Flows and not differing reliably from each other. Human similarity also distinguished the algorithms: Proto was more similar to human-generated CFs than both DiCE and Normalizing Flows,

¹<https://github.com/ukuhl/Hu-CEMNIST>

Table 2: Summary of most important effects. Full omnibus tests, post-hoc contrasts, and correlations are reported in the supplementary materials.

Main result	Statistical evidence
Which methods best match human change maps?	
DiCE and Proto show high human agreement and both outperform Normalizing Flows; they do not differ from each other.	Friedman: $\chi^2(2) = 135.56$, $p < .001$, $W = .753$; DiCE/Proto vs. Normalizing Flows: $r = .868$.
Which methods are most similar to human-generated CFEs?	
Proto shows the highest human similarity, outperforming both DiCE and Normalizing Flows. DiCE and Normalizing Flows do not differ reliably.	Friedman: $\chi^2(2) = 70.07$, $p < .001$, $W = .389$; Proto vs. DiCE/Normalizing Flows: $r = .747-.761$.
Which methods have highest training-set similarity?	
DiCE and Proto obtain the highest training-set-similarity scores and do not differ from each other. Human-generated CFEs are intermediate; Normalizing Flows scores lowest.	Friedman: $\chi^2(3) = 202.89$, $p < .001$, $W = .751$; DiCE/Proto vs. Normalizing Flows: $r \geq .867$; DiCE/Proto vs. humans: $r = .637-.866$.
Which methods change least?	
Human-generated CFEs make the sparsest edits. DiCE and Proto change slightly more, while Normalizing Flows changes nearly the whole image.	Friedman: $\chi^2(3) = 192.17$, $p < .001$, $W = .712$; humans vs. DiCE/Proto: $r = .598-.675$; DiCE/Proto vs. Normalizing Flows: $r = .868$.
What predicts human agreement?	
Higher human agreement is associated with higher human similarity, higher training-set similarity, and smaller changes.	Agreement-similarity: $\rho = .261$, $p < .001$; agreement-training set: $\rho = .687$, $p < .001$; agreement-change: $\rho = -.618$, $p < .001$.

whereas DiCE and Normalizing Flows did not differ reliably. For training-set similarity, DiCE and Proto achieved the highest scores and did not differ from each other. Human-generated CFEs showed intermediate training-set similarity, scoring lower than DiCE and Proto but higher than Normalizing Flows.

For the percentage of change, human-generated CFEs made the sparsest edits, followed by DiCE and Proto, while Normalizing Flows changed nearly the entire image. Across metrics, higher human agreement was associated with higher human similarity, higher training-set similarity, and smaller changes.

These results extend prior observations that human and machine counterfactuals may pursue different transformation goals [Delaney *et al.*, 2023; Qi *et al.*, 2024]. Human alignment captures semantic, stroke-level similarity, but not proximity to the classifier’s decision boundary. Counterfactuals may thus be model-faithful yet visually unintuitive, or intuitive yet farther from the boundary.

5 Summary & Conclusion

Hu-CEMIST provides a benchmark for studying visual CFEs from a human-centered perspective. We collected human-generated counterfactual edits for all ordered non-identical MNIST digit transformations and quality-controlled them using both a minimum-change criterion and independent recognizability ratings. This yields a large set of human-

produced counterfactual digit images that can be used not only to evaluate whether counterfactual methods change a classifier’s decision, but also whether their changes resemble the kinds of visual modifications humans naturally make. Following human-grounded XAI evaluation [Mohseni *et al.*, 2021; Kazmierczak *et al.*, 2026], this comparison is best viewed as a purpose-dependent benchmark dimension that complements classifier validity and boundary fidelity.

Our comparison with DiCE, prototype-guided counterfactuals, and normalizing-flow-based counterfactuals suggests that current algorithmic methods differ systematically from human-generated CFEs. Human edits tend to concentrate on meaningful stroke regions and show greater variation across participants. This heterogeneity may reflect multiple transformation strategies, although confirming such clusters requires dedicated analysis. DiCE and Proto are closer to the target-class MNIST training distribution than human-generated CFEs, whereas Normalizing Flows shows lower training-set similarity. Thus, training-set similarity alone does not imply human-aligned plausibility.

The statistical analyses support this interpretation and show that alignment depends on the metric. DiCE and Proto best matched human change maps, Proto best matched final human-generated CFEs, and DiCE and Proto achieved the highest training-set similarity. Human-generated CFEs were sparsest, whereas Normalizing Flows changed almost the entire image and showed low training-set similarity. Correlations further indicated that higher human agreement was associated with higher human similarity, higher training-set similarity, and smaller changes.

5.1 Limitations

Hu-CEMIST is deliberately restricted to MNIST. This controlled setting makes the benchmark easy to analyze, but it limits generality: handwritten digits are grayscale, low-resolution, and visually simple. Natural-image counterfactuals would require richer editing interfaces and additional quality-control procedures. Moreover, Hu-CEMIST uses one representative source image per digit. Different source exemplars may elicit different human edit strategies, especially for ambiguous or atypical handwriting styles. Future versions should therefore sample multiple source exemplars per digit and analyze how source morphology affects human counterfactual strategies.

A second limitation concerns annotation. Recognition judgments were provided by annotators from one regional writing tradition, and digit-writing conventions vary across cultural contexts. This may affect which edited images are considered recognizable. Future extensions should include annotators from different writing traditions and should explicitly analyze cross-cultural variation in digit recognition.

A third limitation concerns the interpretation of the SSIM-to-training-data metric. Human-generated CFEs showed lower training-set similarity than DiCE and Proto, but higher training-set similarity than Normalizing Flows. On the one hand, this supports the view that humans do not simply reproduce highly training-set-like target digits, but instead make meaningful edits to the given source image. On the other hand, the drawing interface allowed direct pixel-level

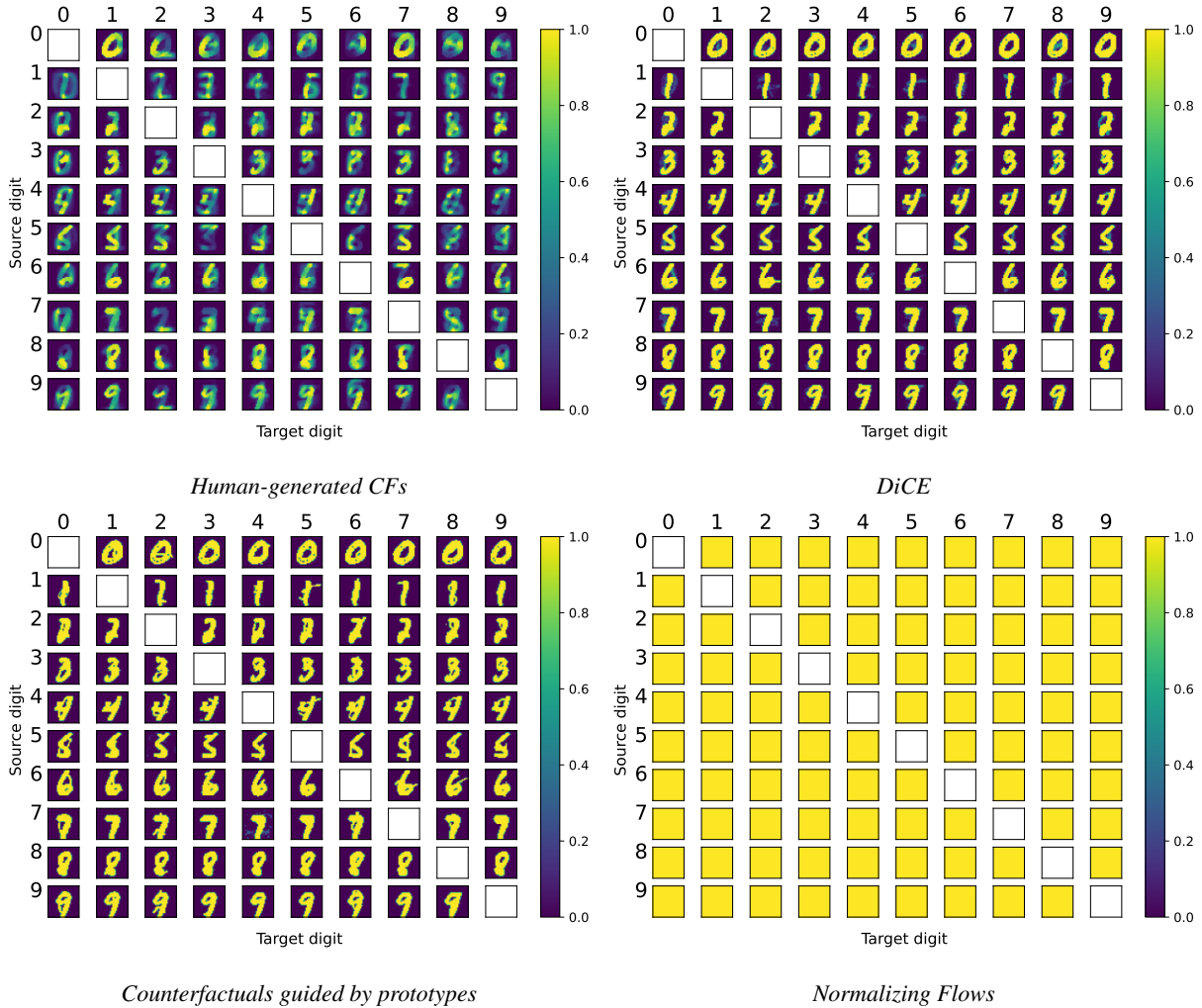


Figure 1: Pixel-wise change frequency for human-generated and algorithmically generated counterfactuals, averaged across all user images / samples per transformation. Brighter pixels indicate locations that were on average changed more consistently across counterfactual instances. For Normalizing Flows, nearly every pixel changes in almost every generated image, producing an approximately uniform full-image change map rather than localized stroke edits.

modifications, which may naturally produce less smooth images than the original MNIST training samples. This may partly disadvantage human-generated CFEs under an SSIM-to-training-data metric. We therefore interpret this metric as training-set similarity, not as a complete measure of visual quality or human plausibility.

Further, the current algorithmic comparison covers three representative counterfactual methods for a single classifier (i.e., MLP) only. The results should therefore be read as evidence of a broader alignment issue rather than as a complete ranking of counterfactual generation approaches.

Finally, a related conceptual limitation is that human agreement is not a universal criterion for CFE quality. HuCEMNIST is most directly relevant when semantic interpretability, communication, or alignment with human concepts is part of the intended use. Future evaluations should therefore report human alignment alongside model-centered criteria in relevant contexts.

5.2 Future Work

In future work, we will extend comparisons to additional CFE methods, particularly generative or plausibility-driven approaches that aim for semantically richer CFEs [Kenny and Keane, 2021; Zhao *et al.*, 2021], as they may align more closely with human-produced edits than proximity-based methods. Second, the benchmark can be extended beyond aggregate similarity metrics. The human data make it possible to model transformation-specific edit distributions, identify common stroke-level strategies, and evaluate whether algorithmic CFEs fall inside or outside the range of human edits. This could support new evaluation metrics based on human change-map likelihood, cluster membership, or transformation-specific edit prototypes. Third, future work should examine the behavioral relevance of human alignment. A natural next step is to test whether CFEs that better match human edit patterns are also more useful to users, for ex-

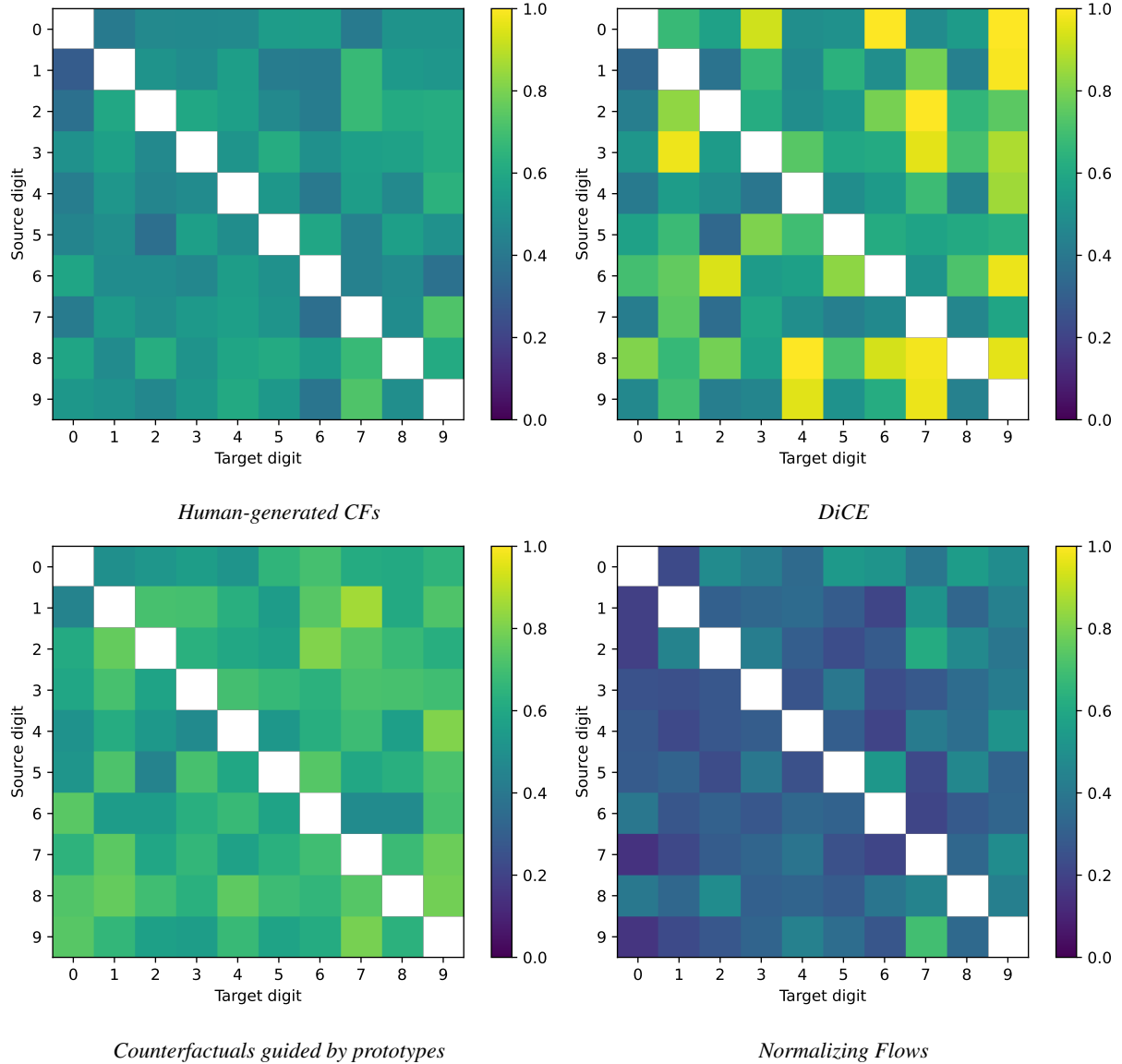


Figure 2: Average training-set similarity (i.e., maximum SSIM score Eq. 3) of counterfactuals with MNIST training data.

ample by improving task understanding, model debugging, trust calibration, or human prediction of model behavior. This would connect the present benchmark to downstream human-centered XAI evaluation. Finally, future work could focus on participant groups by comparing individuals with different levels of machine learning or MNIST familiarity. Experts asked to identify the smallest classifier-effective change may produce edits that differ systematically from those generated under the present instruction.

5.3 Conclusion

Hu-CEMIST introduces a quality-controlled benchmark of human-generated CFEs for MNIST digit transformations. The benchmark shows that human counterfactual edits are indeed sparse, yet not merely classifier-targeted pixel perturbations or replicas of training-set examples. Instead, they re-

flect meaningful visual changes to digit structure and often differ from the outputs of established algorithmic CFE methods. The presented results also show that alignment depends on the metric: DiCE and Proto best match human change maps, Proto best matches final human-generated counterfactual images, and DiCE and Proto are most similar to target-class MNIST training samples. Thus, we conclude that validity, sparsity, proximity, and distributional similarity do not by themselves characterize human-aligned visual change. While human alignment is not relevant to every explanatory purpose, it should be evaluated whenever CFEs are intended to communicate semantically meaningful changes to people. By providing human-generated reference data across all digit transformations, Hu-CEMIST enables systematic evaluation of counterfactual generators as both model-effective and semantically meaningful explanation tools.

Ethical Statement

The study was approved by the Ethics Committee of Bielefeld University, Germany (No. 2025-081).

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Supplementary Materials

These supplementary materials provide additional details and analyses supporting the paper *Hu-CEMNIST: A Benchmark Dataset of Human-Generated Counterfactual Explanations for MNIST*. We first describe the study interface, procedure, and survey items used in the human evaluation. We then report extended quantitative results on the generated human- and algorithmically generated counterfactuals, including their similarity to MNIST training data, the percentage of changed pixels, and, for the algorithmically generated outputs, their overlap and similarity with human-generated counterfactuals. Finally, we provide descriptive statistics, omnibus and post-hoc statistical tests, correlations among evaluation metrics, a summary of significant effects, and visualizations of metric distributions and relationships.

Study Interface, Procedure, and Survey Items

Current Task:
If these changes were applied to the 0, it would become a 5
Amount of change: 0%

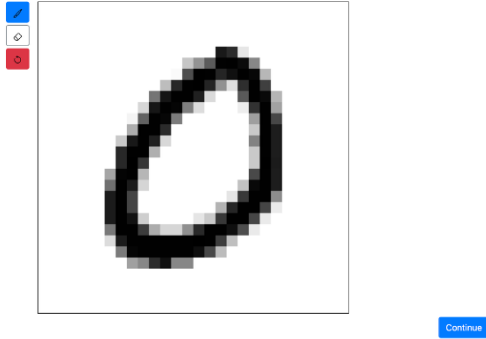


Figure 3: Screenshot of the annotation interface used in the study. Participants edited the source symbol using a brush, eraser, and undo button. The interface also displayed the percentage of changed pixels, allowing participants to monitor whether they had reached the minimum 2% change requirement.

Annotation Interface

Participants completed the counterfactual annotation tasks using a custom web-based drawing interface. The interface resembled a minimal pixel-level paint tool and provided three core functions: a brush for adding pixels, an eraser for removing pixels, and an undo button for reversing the most recent action. The source symbol was displayed on the drawing canvas, and participants were instructed to modify it so that it would be recognizable as the specified target symbol while changing as little as possible.

The interface continuously displayed the percentage of changed pixels. Participants were instructed to reach at least 2% change for each image. This threshold was intended to discourage null or near-null submissions while preserving the emphasis on minimal, meaningful edits.

Procedure

Participants first provided informed consent and then received written instructions explaining the goal of the task, the use of the annotation tool, the minimum-change requirement, and the need to avoid using the browser’s back button. They were informed that they would watch a video tutorial and then complete four comprehension questions before starting the annotation tasks.

The video tutorial introduced the interface and task procedure using an example symbol that was not part of the MNIST digit stimuli, in order to avoid priming participants

toward particular digit-editing strategies. Participants then completed four comprehension questions. Each question allowed two attempts. Participants who failed any question twice were screened out and received partial compensation. The comprehension-check items are shown in Table 3.

After passing the comprehension check, participants completed three annotation tasks. In each task, they were shown a source symbol and a target-symbol label and were asked to minimally edit the source symbol so that it would be recognizable as the target. After the annotation tasks, participants completed demographic and feedback questions. Finally, they received a debriefing explaining that the study aimed to understand how people visually distinguish between digits and to support the evaluation of human- and computer-generated explanations.

Distribution of Collected Annotations

The annotation tasks covered all 90 ordered non-identical source-target digit transformations. Data collection was conducted in 13 waves, with additional annotations collected for transformations that remained underrepresented or yielded fewer valid responses after quality control. Table 4 reports the number of retained annotations for each source-target pair.

Post-Task Survey

After completing the annotation tasks, participants answered demographic, usability, attention-check, and feedback questions. The post-task survey items are shown in Table 5.

No.	Question	Response options
1	What is the main goal of the annotation tool?	(a) To create a new font. (b) To make minimal changes to a symbol to transform it into another symbol. (c) To make as many changes as possible to a symbol to transform it into another symbol. (d) To identify parts of a symbol that make it look appealing.
2	How much change should apply to each image?	(a) Exactly 2%. (b) At least 5%. (c) At least 2%. (d) At least 2 px.
3	Which symbol denotes the “Brush” feature, allowing users to add pixels?	(a) The brush icon. (b) The eraser icon. (c) The undo icon. (d) The tool does not allow users to add any changes.
4	Which statement best describes the “Undo” feature?	(a) It automatically makes all necessary changes. (b) It recalculates the exact percentage of change. (c) It saves all your annotations. (d) It reverses your most recent action.

Table 3: Comprehension-check items shown before the annotation tasks. Participants had two attempts per item. Failing any item twice resulted in screening out and partial compensation.

Source digit	Target digit										Total
	0	1	2	3	4	5	6	7	8	9	
0	–	52	50	50	50	51	50	52	52	51	458
1	55	–	52	53	53	54	50	54	53	50	474
2	53	51	–	50	52	52	52	53	51	50	464
3	50	54	52	–	51	50	54	50	53	53	467
4	50	54	54	50	–	51	50	50	51	51	461
5	51	53	51	54	51	–	52	51	55	50	468
6	55	50	50	50	50	50	–	52	52	51	460
7	52	50	54	51	52	52	55	–	51	51	468
8	52	51	54	51	52	52	54	50	–	53	469
9	50	50	53	54	52	52	52	51	56	–	470
Total	468	465	470	463	463	464	469	463	474	460	4,659

Table 4: Distribution of the 4,659 retained human-generated counterfactual annotations across ordered source-target digit transformations. Rows indicate source digits and columns indicate target digits. Diagonal cells are omitted because the benchmark contains only non-identical transformations.

No.	Question	Response type	Response options
1	Please indicate your age.	Single choice	18–24 years; 25–34 years; 35–44 years; 45–54 years; 55–65 years; 65 years and older; Prefer not to say.
2	Please indicate your gender identity.	Single choice	Male; Female; Non-binary; Transgender; Other; Prefer not to say.
3	The instructions were clear enough to understand how to make minimal changes while reaching at least 2% change.	Five-point Likert scale	Strongly disagree; Disagree; Neutral; Agree; Strongly agree.
4	It was difficult to use the brush, eraser, and undo annotations.	Five-point Likert scale	Strongly disagree; Disagree; Neutral; Agree; Strongly agree.
5	I am confident that my annotations capture the essential differences between two symbols.	Five-point Likert scale	Strongly disagree; Disagree; Neutral; Agree; Strongly agree.
6	Please select “Strongly disagree” to confirm that you are paying attention.	Attention check	Strongly disagree; Disagree; Neutral; Agree; Strongly agree.
7	Did you follow a specific annotation strategy?	Yes/No	Yes; No.
8	If you followed a specific annotation strategy: How did you decide which features to highlight or erase?	Optional short text	Open response.
9	Did the 2% minimal-change requirement affect your annotation strategy?	Yes/No	Yes; No.
10	If the 2% minimal-change requirement affected your annotation strategy, how did it do that?	Optional short text	Open response.
11	Is there anything else you would like to share about your experience or about how we can improve the tool?	Optional short text	Open response.

Table 5: Post-task survey items completed after the three annotation tasks.

Similarity between generated counterfactuals and MNIST training data

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.41 ± 0.06	0.47 ± 0.06	0.48 ± 0.08	0.48 ± 0.09	0.55 ± 0.13	0.56 ± 0.06	0.41 ± 0.09	0.51 ± 0.11	0.51 ± 0.08
	1	0.29 ± 0.08	—	0.52 ± 0.06	0.49 ± 0.08	0.56 ± 0.07	0.42 ± 0.05	0.41 ± 0.06	0.67 ± 0.06	0.54 ± 0.1	0.52 ± 0.08
	2	0.37 ± 0.08	0.6 ± 0.08	—	0.6 ± 0.05	0.57 ± 0.08	0.47 ± 0.1	0.42 ± 0.08	0.67 ± 0.1	0.61 ± 0.07	0.62 ± 0.04
	3	0.51 ± 0.1	0.57 ± 0.08	0.47 ± 0.05	—	0.52 ± 0.09	0.62 ± 0.06	0.5 ± 0.1	0.56 ± 0.09	0.57 ± 0.09	0.62 ± 0.06
	4	0.42 ± 0.1	0.53 ± 0.11	0.46 ± 0.06	0.48 ± 0.08	—	0.53 ± 0.1	0.4 ± 0.05	0.56 ± 0.05	0.47 ± 0.05	0.64 ± 0.1
	5	0.45 ± 0.11	0.49 ± 0.08	0.36 ± 0.07	0.56 ± 0.07	0.49 ± 0.08	—	0.59 ± 0.07	0.44 ± 0.06	0.57 ± 0.1	0.5 ± 0.09
	6	0.59 ± 0.1	0.49 ± 0.07	0.49 ± 0.08	0.46 ± 0.08	0.56 ± 0.08	0.51 ± 0.07	—	0.44 ± 0.06	0.48 ± 0.06	0.37 ± 0.03
	7	0.41 ± 0.07	0.55 ± 0.1	0.49 ± 0.03	0.54 ± 0.04	0.56 ± 0.09	0.52 ± 0.09	0.37 ± 0.05	—	0.48 ± 0.07	0.73 ± 0.07
	8	0.59 ± 0.07	0.48 ± 0.09	0.62 ± 0.06	0.52 ± 0.04	0.59 ± 0.1	0.61 ± 0.07	0.55 ± 0.05	0.67 ± 0.08	—	0.61 ± 0.06
	9	0.53 ± 0.08	0.51 ± 0.05	0.46 ± 0.04	0.53 ± 0.07	0.61 ± 0.07	0.53 ± 0.08	0.39 ± 0.04	0.72 ± 0.06	0.49 ± 0.05	—

Table 6: *Human-generated CFs*: Similarity (i.e., maximum SSIM score Eq. 3) of counterfactuals with MNIST training data. We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.67 ± 0.08	0.57 ± 0.08	0.93 ± 0.19	0.49 ± 0.12	0.51 ± 0.12	1.0 ± 0.0	0.48 ± 0.01	0.55 ± 0.27	1.0 ± 0.0
	1	0.34 ± 0.06	—	0.38 ± 0.05	0.67 ± 0.2	0.47 ± 0.08	0.64 ± 0.07	0.5 ± 0.05	0.79 ± 0.23	0.43 ± 0.14	0.99 ± 0.06
	2	0.43 ± 0.1	0.83 ± 0.1	—	0.62 ± 0.13	0.49 ± 0.08	0.53 ± 0.07	0.8 ± 0.22	1.0 ± 0.0	0.66 ± 0.08	0.75 ± 0.25
	3	0.52 ± 0.18	0.97 ± 0.05	0.55 ± 0.2	—	0.74 ± 0.23	0.6 ± 0.16	0.61 ± 0.1	0.96 ± 0.1	0.71 ± 0.22	0.88 ± 0.19
	4	0.41 ± 0.07	0.55 ± 0.08	0.5 ± 0.09	0.39 ± 0.16	—	0.49 ± 0.08	0.54 ± 0.16	0.69 ± 0.15	0.45 ± 0.06	0.86 ± 0.15
	5	0.57 ± 0.11	0.68 ± 0.04	0.34 ± 0.04	0.81 ± 0.16	0.69 ± 0.22	—	0.62 ± 0.11	0.58 ± 0.1	0.61 ± 0.14	0.63 ± 0.16
	6	0.7 ± 0.21	0.76 ± 0.07	0.94 ± 0.14	0.55 ± 0.08	0.57 ± 0.08	0.83 ± 0.23	—	0.52 ± 0.06	0.72 ± 0.22	0.97 ± 0.1
	7	0.42 ± 0.08	0.75 ± 0.04	0.36 ± 0.07	0.59 ± 0.11	0.49 ± 0.06	0.43 ± 0.04	0.48 ± 0.05	—	0.46 ± 0.06	0.59 ± 0.08
	8	0.81 ± 0.17	0.67 ± 0.06	0.79 ± 0.19	0.57 ± 0.16	1.0 ± 0.0	0.71 ± 0.19	0.94 ± 0.12	0.98 ± 0.07	—	0.96 ± 0.07
	9	0.47 ± 0.08	0.7 ± 0.03	0.43 ± 0.07	0.46 ± 0.17	0.96 ± 0.13	0.51 ± 0.07	0.62 ± 0.24	0.97 ± 0.08	0.44 ± 0.05	—

Table 7: *DiCE*: Similarity (i.e., maximum SSIM score Eq. 3) of counterfactuals with MNIST training data. We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.5 ± 0.0	0.53 ± 0.0	0.56 ± 0.0	0.52 ± 0.0	0.65 ± 0.0	0.7 ± 0.0	0.62 ± 0.0	0.61 ± 0.0	0.65 ± 0.0
	1	0.45 ± 0.01	—	0.71 ± 0.0	0.7 ± 0.0	0.63 ± 0.0	0.55 ± 0.0	0.74 ± 0.0	0.86 ± 0.0	0.61 ± 0.0	0.72 ± 0.0
	2	0.61 ± 0.0	0.76 ± 0.0	—	0.64 ± 0.0	0.6 ± 0.0	0.57 ± 0.0	0.81 ± 0.0	0.73 ± 0.0	0.67 ± 0.0	0.63 ± 0.0
	3	0.59 ± 0.0	0.71 ± 0.0	0.58 ± 0.0	—	0.7 ± 0.0	0.67 ± 0.0	0.64 ± 0.0	0.71 ± 0.0	0.71 ± 0.0	0.69 ± 0.0
	4	0.51 ± 0.0	0.62 ± 0.0	0.54 ± 0.0	0.48 ± 0.0	—	0.53 ± 0.0	0.63 ± 0.0	0.68 ± 0.0	0.56 ± 0.0	0.81 ± 0.0
	5	0.52 ± 0.0	0.72 ± 0.0	0.45 ± 0.01	0.71 ± 0.0	0.6 ± 0.0	—	0.74 ± 0.0	0.6 ± 0.0	0.63 ± 0.0	0.72 ± 0.0
	6	0.74 ± 0.0	0.55 ± 0.0	0.55 ± 0.0	0.63 ± 0.0	0.67 ± 0.0	0.58 ± 0.0	—	0.48 ± 0.0	0.48 ± 0.0	0.71 ± 0.0
	7	0.64 ± 0.0	0.75 ± 0.0	0.59 ± 0.0	0.66 ± 0.0	0.57 ± 0.06	0.63 ± 0.0	0.69 ± 0.0	—	0.68 ± 0.0	0.77 ± 0.0
	8	0.73 ± 0.0	0.77 ± 0.0	0.69 ± 0.0	0.64 ± 0.0	0.75 ± 0.0	0.69 ± 0.0	0.66 ± 0.0	0.74 ± 0.0	—	0.79 ± 0.0
	9	0.74 ± 0.0	0.66 ± 0.0	0.57 ± 0.0	0.56 ± 0.0	0.67 ± 0.0	0.58 ± 0.0	0.61 ± 0.0	0.8 ± 0.0	0.64 ± 0.0	—

Table 8: *Counterfactuals guided by prototypes*: Similarity (i.e., maximum SSIM score Eq. 3) of counterfactuals with MNIST training data. We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
Source digit		0	1	2	3	4	5	6	7	8	9
	0	—	0.22 ± 0.0	0.48 ± 0.0	0.42 ± 0.0	0.34 ± 0.0	0.54 ± 0.0	0.52 ± 0.0	0.39 ± 0.0	0.55 ± 0.0	0.49 ± 0.0
	1	0.19 ± 0.0	—	0.31 ± 0.0	0.33 ± 0.0	0.35 ± 0.0	0.29 ± 0.0	0.21 ± 0.0	0.51 ± 0.0	0.33 ± 0.0	0.44 ± 0.0
	2	0.19 ± 0.0	0.45 ± 0.0	—	0.42 ± 0.0	0.3 ± 0.0	0.24 ± 0.0	0.29 ± 0.0	0.61 ± 0.0	0.47 ± 0.0	0.39 ± 0.0
	3	0.25 ± 0.0	0.25 ± 0.0	0.27 ± 0.0	—	0.26 ± 0.0	0.4 ± 0.0	0.24 ± 0.0	0.28 ± 0.0	0.36 ± 0.0	0.42 ± 0.0
	4	0.28 ± 0.0	0.22 ± 0.0	0.27 ± 0.0	0.3 ± 0.0	—	0.3 ± 0.0	0.21 ± 0.0	0.42 ± 0.0	0.36 ± 0.0	0.51 ± 0.0
	5	0.29 ± 0.0	0.32 ± 0.0	0.23 ± 0.0	0.4 ± 0.0	0.25 ± 0.0	—	0.53 ± 0.0	0.22 ± 0.0	0.46 ± 0.0	0.32 ± 0.0
	6	0.39 ± 0.0	0.27 ± 0.0	0.31 ± 0.0	0.27 ± 0.0	0.33 ± 0.0	0.31 ± 0.0	—	0.21 ± 0.0	0.28 ± 0.0	0.32 ± 0.0
	7	0.15 ± 0.0	0.22 ± 0.0	0.29 ± 0.0	0.33 ± 0.0	0.39 ± 0.0	0.25 ± 0.0	0.21 ± 0.0	—	0.33 ± 0.0	0.49 ± 0.0
	8	0.4 ± 0.0	0.33 ± 0.0	0.49 ± 0.0	0.31 ± 0.0	0.32 ± 0.0	0.29 ± 0.0	0.39 ± 0.0	0.45 ± 0.0	—	0.43 ± 0.0
	9	0.15 ± 0.0	0.22 ± 0.0	0.27 ± 0.0	0.32 ± 0.0	0.44 ± 0.0	0.36 ± 0.0	0.27 ± 0.0	0.7 ± 0.0	0.34 ± 0.0	—

Table 9: *Normalizing Flows*: Similarity (i.e., maximum SSIM score Eq. 3) of counterfactuals with MNIST training data. We report the mean and standard deviations (over all trials) rounded to two decimal places.

Percentage of change in the generated counterfactuals

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.31 ± 0.03	0.24 ± 0.07	0.25 ± 0.06	0.27 ± 0.05	0.26 ± 0.06	0.17 ± 0.06	0.32 ± 0.04	0.23 ± 0.08	0.18 ± 0.06
	1	0.2 ± 0.05	—	0.16 ± 0.05	0.19 ± 0.04	0.14 ± 0.04	0.2 ± 0.04	0.16 ± 0.05	0.12 ± 0.04	0.22 ± 0.06	0.15 ± 0.05
	2	0.21 ± 0.06	0.18 ± 0.03	—	0.17 ± 0.04	0.22 ± 0.04	0.23 ± 0.03	0.25 ± 0.04	0.14 ± 0.05	0.21 ± 0.05	0.18 ± 0.03
	3	0.22 ± 0.07	0.22 ± 0.02	0.17 ± 0.04	—	0.24 ± 0.03	0.15 ± 0.07	0.19 ± 0.05	0.2 ± 0.04	0.14 ± 0.06	0.14 ± 0.06
	4	0.23 ± 0.06	0.18 ± 0.07	0.22 ± 0.03	0.21 ± 0.06	—	0.21 ± 0.06	0.22 ± 0.03	0.15 ± 0.04	0.19 ± 0.05	0.14 ± 0.07
	5	0.2 ± 0.06	0.21 ± 0.03	0.2 ± 0.04	0.12 ± 0.05	0.19 ± 0.05	—	0.11 ± 0.04	0.21 ± 0.04	0.17 ± 0.06	0.17 ± 0.07
	6	0.18 ± 0.05	0.21 ± 0.04	0.22 ± 0.06	0.23 ± 0.03	0.22 ± 0.04	0.22 ± 0.06	—	0.21 ± 0.04	0.22 ± 0.05	0.23 ± 0.03
	7	0.21 ± 0.07	0.19 ± 0.06	0.13 ± 0.05	0.18 ± 0.06	0.2 ± 0.05	0.23 ± 0.04	0.23 ± 0.05	—	0.21 ± 0.07	0.15 ± 0.07
	8	0.13 ± 0.06	0.24 ± 0.03	0.15 ± 0.04	0.16 ± 0.04	0.26 ± 0.04	0.23 ± 0.04	0.19 ± 0.05	0.2 ± 0.04	—	0.18 ± 0.06
	9	0.18 ± 0.06	0.19 ± 0.06	0.14 ± 0.05	0.19 ± 0.06	0.19 ± 0.06	0.22 ± 0.05	0.22 ± 0.04	0.14 ± 0.05	0.17 ± 0.06	—

Table 10: *Human-generated CFs*: Percentage of change (Eq. 2). We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.33 ± 0.01	0.33 ± 0.0	0.33 ± 0.0	0.33 ± 0.01	0.33 ± 0.01	0.33 ± 0.0	0.32 ± 0.0	0.34 ± 0.01	0.34 ± 0.0
	1	0.16 ± 0.01	—	0.15 ± 0.01	0.15 ± 0.01	0.14 ± 0.01	0.14 ± 0.01	0.14 ± 0.01	0.14 ± 0.01	0.15 ± 0.01	0.16 ± 0.0
	2	0.22 ± 0.01	0.2 ± 0.0	—	0.21 ± 0.01	0.21 ± 0.0	0.21 ± 0.01	0.21 ± 0.01	0.2 ± 0.0	0.21 ± 0.0	0.22 ± 0.02
	3	0.23 ± 0.01	0.22 ± 0.0	0.23 ± 0.01	—	0.23 ± 0.0	0.23 ± 0.01	0.23 ± 0.01	0.23 ± 0.01	0.24 ± 0.01	0.23 ± 0.0
	4	0.22 ± 0.01	0.21 ± 0.0	0.22 ± 0.0	0.22 ± 0.01	—	0.21 ± 0.01	0.22 ± 0.01	0.21 ± 0.01	0.21 ± 0.01	0.21 ± 0.01
	5	0.21 ± 0.01	0.19 ± 0.0	0.2 ± 0.01	0.2 ± 0.01	0.2 ± 0.01	—	0.2 ± 0.0	0.2 ± 0.01	0.2 ± 0.01	0.2 ± 0.01
	6	0.24 ± 0.01	0.23 ± 0.0	0.25 ± 0.0	0.24 ± 0.01	0.24 ± 0.0	0.25 ± 0.01	—	0.23 ± 0.0	0.25 ± 0.01	0.24 ± 0.0
	7	0.23 ± 0.01	0.21 ± 0.0	0.23 ± 0.01	0.22 ± 0.0	0.22 ± 0.0	0.22 ± 0.0	0.22 ± 0.0	—	0.22 ± 0.0	0.22 ± 0.0
	8	0.25 ± 0.0	0.24 ± 0.0	0.24 ± 0.01	0.24 ± 0.01	0.24 ± 0.0	0.25 ± 0.0	0.25 ± 0.0	0.24 ± 0.0	—	0.24 ± 0.0
	9	0.22 ± 0.01	0.21 ± 0.0	0.22 ± 0.01	0.22 ± 0.0	0.22 ± 0.0	0.22 ± 0.0	0.23 ± 0.01	0.21 ± 0.0	0.22 ± 0.0	—

Table 11: *DiCE*: Percentage of change (Eq. 2). We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.34 ± 0.0	0.33 ± 0.0	0.32 ± 0.0	0.33 ± 0.0	0.32 ± 0.0	0.33 ± 0.0	0.33 ± 0.0	0.34 ± 0.0	0.32 ± 0.0
	1	0.15 ± 0.0	—	0.16 ± 0.0	0.14 ± 0.0	0.15 ± 0.0	0.17 ± 0.0	0.15 ± 0.0	0.14 ± 0.0	0.16 ± 0.0	0.15 ± 0.0
	2	0.22 ± 0.0	0.2 ± 0.0	—	0.21 ± 0.0	0.22 ± 0.0	0.22 ± 0.0	0.22 ± 0.0	0.19 ± 0.0	0.21 ± 0.0	0.21 ± 0.0
	3	0.23 ± 0.0	0.22 ± 0.0	0.23 ± 0.0	—	0.23 ± 0.0	0.22 ± 0.0	0.23 ± 0.0	0.22 ± 0.0	0.23 ± 0.0	0.22 ± 0.0
	4	0.24 ± 0.0	0.22 ± 0.0	0.23 ± 0.0	0.23 ± 0.0	—	0.23 ± 0.0	0.23 ± 0.0	0.21 ± 0.0	0.22 ± 0.0	0.21 ± 0.0
	5	0.22 ± 0.0	0.21 ± 0.0	0.21 ± 0.01	0.2 ± 0.0	0.2 ± 0.0	—	0.2 ± 0.0	0.21 ± 0.0	0.2 ± 0.0	0.21 ± 0.0
	6	0.25 ± 0.0	0.24 ± 0.0	0.24 ± 0.0	0.25 ± 0.0	0.24 ± 0.0	0.23 ± 0.0	—	0.25 ± 0.0	0.26 ± 0.0	0.24 ± 0.0
	7	0.24 ± 0.0	0.21 ± 0.0	0.24 ± 0.0	0.23 ± 0.0	0.24 ± 0.04	0.22 ± 0.0	0.24 ± 0.0	—	0.23 ± 0.0	0.22 ± 0.0
	8	0.24 ± 0.0	0.24 ± 0.0	0.24 ± 0.0	0.24 ± 0.0	0.23 ± 0.0	0.24 ± 0.0	0.25 ± 0.0	0.23 ± 0.0	—	0.24 ± 0.0
	9	0.24 ± 0.0	0.22 ± 0.0	0.24 ± 0.0	0.22 ± 0.0	0.21 ± 0.0	0.21 ± 0.0	0.24 ± 0.0	0.21 ± 0.0	0.24 ± 0.0	—

Table 12: *Counterfactuals guided by prototypes*: Percentage of change (Eq. 2). We report the mean and standard deviations (over all trials) rounded to two decimal places.

	Target digit										
	0	1	2	3	4	5	6	7	8	9	
Source digit	0	—	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
	1	1.0 ± 0.0	—	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
	2	1.0 ± 0.0	1.0 ± 0.0	—	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
	3	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	—	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
	4	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	—	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
	5	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	—	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
	6	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	—	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
	7	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	—	1.0 ± 0.0	1.0 ± 0.0
	8	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	—	1.0 ± 0.0
	9	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	—

Table 13: *Normalizing Flows*: Percentage of change (Eq. 2). We report the mean and standard deviations (over all trials) rounded to two decimal places.

Overlap between algorithmically generated and human-generated counterfactuals

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.96 ± 0.02	0.85 ± 0.01	0.87 ± 0.0	0.87 ± 0.01	0.9 ± 0.01	0.7 ± 0.0	0.97 ± 0.0	0.89 ± 0.01	0.84 ± 0.0
	1	0.67 ± 0.02	—	0.65 ± 0.02	0.83 ± 0.05	0.77 ± 0.02	0.59 ± 0.02	0.63 ± 0.02	0.87 ± 0.03	0.62 ± 0.02	0.78 ± 0.01
	2	0.88 ± 0.03	0.99 ± 0.01	—	0.78 ± 0.01	0.85 ± 0.01	0.83 ± 0.02	0.79 ± 0.01	0.94 ± 0.0	0.81 ± 0.01	0.8 ± 0.03
	3	0.77 ± 0.02	0.96 ± 0.0	0.87 ± 0.03	—	0.9 ± 0.02	0.84 ± 0.01	0.72 ± 0.02	0.9 ± 0.02	0.78 ± 0.02	0.8 ± 0.01
	4	0.85 ± 0.02	0.95 ± 0.02	0.83 ± 0.02	0.79 ± 0.01	—	0.81 ± 0.02	0.75 ± 0.02	0.73 ± 0.01	0.72 ± 0.01	0.88 ± 0.02
	5	0.84 ± 0.02	0.96 ± 0.01	0.76 ± 0.01	0.81 ± 0.01	0.8 ± 0.01	—	0.72 ± 0.02	0.79 ± 0.01	0.73 ± 0.01	0.81 ± 0.01
	6	0.87 ± 0.02	0.96 ± 0.01	0.74 ± 0.01	0.86 ± 0.02	0.89 ± 0.02	0.83 ± 0.01	—	0.87 ± 0.01	0.74 ± 0.02	0.79 ± 0.0
	7	0.82 ± 0.02	0.98 ± 0.0	0.84 ± 0.02	0.8 ± 0.01	0.74 ± 0.01	0.81 ± 0.01	0.71 ± 0.01	—	0.72 ± 0.01	0.82 ± 0.01
	8	0.88 ± 0.01	0.96 ± 0.01	0.82 ± 0.02	0.77 ± 0.01	0.91 ± 0.0	0.9 ± 0.01	0.85 ± 0.01	0.94 ± 0.0	—	0.84 ± 0.0
	9	0.87 ± 0.01	0.98 ± 0.01	0.83 ± 0.01	0.86 ± 0.01	0.86 ± 0.01	0.89 ± 0.01	0.78 ± 0.03	0.94 ± 0.0	0.83 ± 0.01	—

Table 14: *DiCE*: Overlap in percentage with human-generated counterfactuals (Eq. 5). We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.92 ± 0.0	0.83 ± 0.0	0.86 ± 0.0	0.85 ± 0.0	0.89 ± 0.0	0.7 ± 0.0	0.94 ± 0.0	0.92 ± 0.0	0.89 ± 0.0
	1	0.72 ± 0.01	—	0.66 ± 0.0	0.87 ± 0.0	0.8 ± 0.0	0.55 ± 0.0	0.62 ± 0.0	0.92 ± 0.0	0.71 ± 0.0	0.77 ± 0.0
	2	0.88 ± 0.0	0.99 ± 0.0	—	0.8 ± 0.0	0.93 ± 0.0	0.89 ± 0.0	0.76 ± 0.0	0.98 ± 0.0	0.89 ± 0.0	0.88 ± 0.0
	3	0.78 ± 0.0	0.96 ± 0.0	0.89 ± 0.0	—	0.88 ± 0.0	0.85 ± 0.0	0.7 ± 0.0	0.92 ± 0.0	0.78 ± 0.0	0.8 ± 0.0
	4	0.82 ± 0.0	0.91 ± 0.0	0.78 ± 0.0	0.78 ± 0.0	—	0.83 ± 0.0	0.77 ± 0.0	0.72 ± 0.0	0.69 ± 0.0	0.86 ± 0.0
	5	0.84 ± 0.0	0.91 ± 0.0	0.81 ± 0.03	0.8 ± 0.0	0.82 ± 0.0	—	0.77 ± 0.0	0.81 ± 0.0	0.72 ± 0.0	0.78 ± 0.0
	6	0.86 ± 0.0	0.92 ± 0.0	0.79 ± 0.0	0.83 ± 0.0	0.9 ± 0.0	0.87 ± 0.0	—	0.86 ± 0.0	0.76 ± 0.0	0.81 ± 0.0
	7	0.87 ± 0.0	0.98 ± 0.0	0.79 ± 0.0	0.79 ± 0.0	0.73 ± 0.08	0.78 ± 0.0	0.74 ± 0.0	—	0.74 ± 0.0	0.83 ± 0.0
	8	0.91 ± 0.0	0.95 ± 0.0	0.84 ± 0.0	0.77 ± 0.0	0.91 ± 0.0	0.9 ± 0.0	0.88 ± 0.0	0.97 ± 0.0	—	0.85 ± 0.0
	9	0.9 ± 0.0	0.95 ± 0.0	0.86 ± 0.0	0.86 ± 0.0	0.92 ± 0.01	0.93 ± 0.0	0.76 ± 0.0	0.94 ± 0.0	0.8 ± 0.0	—

Table 15: *Counterfactuals guided by prototypes*: Overlap in percentage with human-generated counterfactuals (Eq. 5). We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.38 ± 0.0	0.46 ± 0.0	0.4 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	0.37 ± 0.0	0.44 ± 0.0	0.41 ± 0.0	0.33 ± 0.0
	1	0.51 ± 0.0	—	0.46 ± 0.0	0.4 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	0.37 ± 0.0	0.44 ± 0.0	0.41 ± 0.0	0.33 ± 0.0
	2	0.51 ± 0.0	0.38 ± 0.0	—	0.4 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	0.37 ± 0.0	0.44 ± 0.0	0.41 ± 0.0	0.33 ± 0.0
	3	0.51 ± 0.0	0.38 ± 0.0	0.46 ± 0.0	—	0.39 ± 0.0	0.49 ± 0.0	0.37 ± 0.0	0.44 ± 0.0	0.41 ± 0.0	0.33 ± 0.0
	4	0.51 ± 0.0	0.38 ± 0.0	0.46 ± 0.0	0.4 ± 0.0	—	0.49 ± 0.0	0.37 ± 0.0	0.44 ± 0.0	0.41 ± 0.0	0.33 ± 0.0
	5	0.51 ± 0.0	0.38 ± 0.0	0.46 ± 0.0	0.4 ± 0.0	0.39 ± 0.0	—	0.37 ± 0.0	0.44 ± 0.0	0.41 ± 0.0	0.33 ± 0.0
	6	0.51 ± 0.0	0.38 ± 0.0	0.46 ± 0.0	0.4 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	—	0.44 ± 0.0	0.41 ± 0.0	0.33 ± 0.0
	7	0.51 ± 0.0	0.38 ± 0.0	0.46 ± 0.0	0.4 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	0.37 ± 0.0	—	0.41 ± 0.0	0.33 ± 0.0
	8	0.51 ± 0.0	0.38 ± 0.0	0.46 ± 0.0	0.4 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	0.37 ± 0.0	0.44 ± 0.0	—	0.33 ± 0.0
	9	0.51 ± 0.0	0.38 ± 0.0	0.46 ± 0.0	0.4 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	0.37 ± 0.0	0.44 ± 0.0	0.41 ± 0.0	—

Table 16: *Normalizing Flows*: Overlap in percentage with human-generated counterfactuals (Eq. 5). We report the mean and standard deviations (over all trials) rounded to two decimal places.

Similarity between algorithmically generated and human-generated counterfactuals

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.6 ± 0.05	0.47 ± 0.06	0.46 ± 0.04	0.36 ± 0.04	0.43 ± 0.07	0.53 ± 0.0	0.41 ± 0.01	0.41 ± 0.11	0.49 ± 0.0
	1	0.33 ± 0.04	—	0.38 ± 0.04	0.51 ± 0.04	0.47 ± 0.04	0.45 ± 0.03	0.43 ± 0.04	0.61 ± 0.12	0.38 ± 0.06	0.7 ± 0.02
	2	0.36 ± 0.05	0.68 ± 0.03	—	0.53 ± 0.05	0.48 ± 0.04	0.4 ± 0.03	0.48 ± 0.03	0.75 ± 0.0	0.54 ± 0.06	0.49 ± 0.04
	3	0.44 ± 0.1	0.65 ± 0.01	0.41 ± 0.04	—	0.56 ± 0.09	0.52 ± 0.07	0.51 ± 0.05	0.59 ± 0.01	0.54 ± 0.08	0.58 ± 0.05
	4	0.39 ± 0.05	0.54 ± 0.05	0.39 ± 0.04	0.38 ± 0.08	—	0.43 ± 0.04	0.43 ± 0.04	0.58 ± 0.13	0.38 ± 0.04	0.69 ± 0.06
	5	0.5 ± 0.08	0.64 ± 0.03	0.37 ± 0.04	0.55 ± 0.04	0.58 ± 0.08	—	0.55 ± 0.07	0.52 ± 0.05	0.5 ± 0.06	0.57 ± 0.05
	6	0.54 ± 0.1	0.65 ± 0.04	0.51 ± 0.05	0.43 ± 0.03	0.51 ± 0.07	0.57 ± 0.1	—	0.5 ± 0.03	0.47 ± 0.07	0.65 ± 0.04
	7	0.4 ± 0.05	0.7 ± 0.03	0.31 ± 0.03	0.49 ± 0.05	0.48 ± 0.04	0.42 ± 0.04	0.41 ± 0.03	—	0.41 ± 0.03	0.53 ± 0.05
	8	0.64 ± 0.08	0.55 ± 0.06	0.63 ± 0.07	0.48 ± 0.06	0.56 ± 0.0	0.57 ± 0.07	0.61 ± 0.05	0.7 ± 0.03	—	0.65 ± 0.02
	9	0.45 ± 0.05	0.68 ± 0.02	0.42 ± 0.04	0.41 ± 0.08	0.62 ± 0.02	0.44 ± 0.03	0.45 ± 0.05	0.8 ± 0.04	0.4 ± 0.03	—

Table 17: *DiCE*: Similarity (i.e., max SSMI score) with human-generated counterfactuals (Eq. 6). We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.41 ± 0.0	0.6 ± 0.0	0.62 ± 0.0	0.61 ± 0.0	0.71 ± 0.0	0.77 ± 0.0	0.39 ± 0.0	0.49 ± 0.0	0.79 ± 0.0
	1	0.49 ± 0.0	—	0.72 ± 0.0	0.69 ± 0.0	0.67 ± 0.0	0.51 ± 0.0	0.61 ± 0.0	0.86 ± 0.0	0.77 ± 0.0	0.7 ± 0.0
	2	0.68 ± 0.0	0.65 ± 0.0	—	0.79 ± 0.0	0.71 ± 0.0	0.69 ± 0.0	0.53 ± 0.0	0.89 ± 0.0	0.78 ± 0.0	0.81 ± 0.0
	3	0.61 ± 0.0	0.61 ± 0.0	0.46 ± 0.0	—	0.64 ± 0.0	0.87 ± 0.0	0.6 ± 0.0	0.59 ± 0.0	0.7 ± 0.0	0.68 ± 0.0
	4	0.56 ± 0.0	0.58 ± 0.0	0.39 ± 0.0	0.54 ± 0.0	—	0.57 ± 0.0	0.49 ± 0.0	0.67 ± 0.0	0.66 ± 0.0	0.88 ± 0.0
	5	0.53 ± 0.0	0.57 ± 0.0	0.45 ± 0.01	0.66 ± 0.0	0.66 ± 0.0	—	0.93 ± 0.0	0.52 ± 0.0	0.69 ± 0.0	0.54 ± 0.0
	6	0.8 ± 0.0	0.59 ± 0.0	0.56 ± 0.0	0.63 ± 0.0	0.59 ± 0.0	0.61 ± 0.0	—	0.42 ± 0.0	0.6 ± 0.0	0.56 ± 0.0
	7	0.61 ± 0.0	0.7 ± 0.0	0.64 ± 0.0	0.7 ± 0.0	0.6 ± 0.07	0.65 ± 0.0	0.46 ± 0.0	—	0.61 ± 0.0	0.89 ± 0.0
	8	0.86 ± 0.0	0.54 ± 0.0	0.84 ± 0.0	0.73 ± 0.0	0.67 ± 0.0	0.68 ± 0.0	0.57 ± 0.0	0.82 ± 0.0	—	0.8 ± 0.0
	9	0.78 ± 0.0	0.58 ± 0.0	0.65 ± 0.0	0.72 ± 0.0	0.78 ± 0.0	0.68 ± 0.0	0.53 ± 0.0	0.88 ± 0.0	0.57 ± 0.0	—

Table 18: *Counterfactuals guided by prototypes*: Similarity (i.e., max SSMI score) with human-generated counterfactuals (Eq. 6). We report the mean and standard deviations (over all trials) rounded to two decimal places.

		Target digit									
		0	1	2	3	4	5	6	7	8	9
Source digit	0	—	0.4 ± 0.0	0.66 ± 0.0	0.72 ± 0.0	0.59 ± 0.0	0.69 ± 0.0	0.78 ± 0.0	0.48 ± 0.0	0.76 ± 0.0	0.85 ± 0.0
	1	0.36 ± 0.0	—	0.5 ± 0.0	0.52 ± 0.0	0.52 ± 0.0	0.48 ± 0.0	0.47 ± 0.0	0.63 ± 0.0	0.47 ± 0.0	0.57 ± 0.0
	2	0.48 ± 0.0	0.47 ± 0.0	—	0.68 ± 0.0	0.49 ± 0.0	0.38 ± 0.0	0.49 ± 0.0	0.81 ± 0.0	0.68 ± 0.0	0.55 ± 0.0
	3	0.46 ± 0.0	0.34 ± 0.0	0.44 ± 0.0	—	0.39 ± 0.0	0.62 ± 0.0	0.46 ± 0.0	0.42 ± 0.0	0.58 ± 0.0	0.66 ± 0.0
	4	0.5 ± 0.0	0.43 ± 0.0	0.41 ± 0.0	0.62 ± 0.0	—	0.42 ± 0.0	0.42 ± 0.0	0.55 ± 0.0	0.59 ± 0.0	0.66 ± 0.0
	5	0.38 ± 0.0	0.4 ± 0.0	0.44 ± 0.0	0.68 ± 0.0	0.49 ± 0.0	—	0.73 ± 0.0	0.28 ± 0.0	0.6 ± 0.0	0.61 ± 0.0
	6	0.53 ± 0.0	0.35 ± 0.0	0.52 ± 0.0	0.62 ± 0.0	0.49 ± 0.0	0.6 ± 0.0	—	0.38 ± 0.0	0.53 ± 0.0	0.33 ± 0.0
	7	0.29 ± 0.0	0.23 ± 0.0	0.32 ± 0.0	0.5 ± 0.0	0.5 ± 0.0	0.5 ± 0.0	0.44 ± 0.0	—	0.54 ± 0.0	0.6 ± 0.0
	8	0.62 ± 0.0	0.37 ± 0.0	0.74 ± 0.0	0.63 ± 0.0	0.49 ± 0.0	0.44 ± 0.0	0.62 ± 0.0	0.58 ± 0.0	—	0.61 ± 0.0
	9	0.26 ± 0.0	0.18 ± 0.0	0.28 ± 0.0	0.51 ± 0.0	0.57 ± 0.0	0.58 ± 0.0	0.43 ± 0.0	0.85 ± 0.0	0.61 ± 0.0	—

Table 19: *Normalizing Flows*: Similarity (i.e., max SSMI score) with human-generated counterfactuals (Eq. 6). We report the mean and standard deviations (over all trials) rounded to two decimal places.

Descriptive Statistics by Metric and Group.

Table 20: Descriptive statistics for each metric and group.

Metric	Group	n	Mean	Median	SD
Human agreement	DiCE	90	0.828	0.834	0.086
Human agreement	Normalizing Flows	90	0.418	0.405	0.054
Human agreement	Proto	90	0.835	0.845	0.085
Human similarity	DiCE	90	0.510	0.502	0.105
Human similarity	Normalizing Flows	90	0.519	0.504	0.138
Human similarity	Proto	90	0.650	0.644	0.124
Training-set similarity	Human-generated	90	0.519	0.515	0.082
Training-set similarity	DiCE	90	0.651	0.613	0.190
Training-set similarity	Normalizing Flows	90	0.341	0.321	0.107
Training-set similarity	Proto	90	0.645	0.644	0.089
Percentage of change	Human-generated	90	0.196	0.197	0.040
Percentage of change	DiCE	90	0.226	0.220	0.044
Percentage of change	Normalizing Flows	90	1.000	1.000	0.000
Percentage of change	Proto	90	0.229	0.228	0.043

Friedman Omnibus Tests.

Table 21: Friedman repeated-measures tests by metric. Kendall's W is reported as the omnibus effect size. Significant effects are shown in bold.

Metric	Groups included	n	χ^2	df	Kendall's W	p
Human agreement	DiCE, Normalizing Flows, Proto	90	135.56	2	0.753	< . 001
Human similarity	DiCE, Normalizing Flows, Proto	90	70.07	2	0.389	< . 001
Training-set similarity	Human-generated, DiCE, Normalizing Flows, Proto	90	202.89	3	0.751	< . 001
Percentage of change	Human-generated, DiCE, Normalizing Flows, Proto	90	192.17	3	0.712	< . 001

Holm-Corrected Post-hoc Comparisons.

Table 22: Paired Wilcoxon signed-rank post-hoc tests with Holm-corrected p -values and effect sizes r . Significant comparisons are shown in bold.

Metric	Contrast	p_{Holm}	r	Effect size	Interpretation
Human agreement	DiCE vs. Normalizing Flows	< .001	0.868	Large	DiCE > Normalizing Flows
Human agreement	DiCE vs. Proto	.072	0.190	Small	No reliable difference
Human agreement	Normalizing Flows vs. Proto	< .001	0.868	Large	Proto > Normalizing Flows
Human similarity	DiCE vs. Normalizing Flows	.205	0.134	Small	No reliable difference
Human similarity	DiCE vs. Proto	< .001	0.761	Large	Proto > DiCE
Human similarity	Normalizing Flows vs. Proto	< .001	0.747	Large	Proto > Normalizing Flows
Training-set similarity	DiCE vs. Human-generated	< .001	0.637	Large	DiCE > Human-generated
Training-set similarity	DiCE vs. Normalizing Flows	< .001	0.867	Large	DiCE > Normalizing Flows
Training-set similarity	DiCE vs. Proto	.757	0.033	Negligible	No reliable difference
Training-set similarity	Human-generated vs. Normalizing Flows	< .001	0.865	Large	Human-generated > Normalizing Flows
Training-set similarity	Human-generated vs. Proto	< .001	0.866	Large	Proto > Human-generated
Training-set similarity	Normalizing Flows vs. Proto	< .001	0.868	Large	Proto > Normalizing Flows
Percentage of change	DiCE vs. Human-generated	< .001	0.598	Large	DiCE > Human-generated
Percentage of change	DiCE vs. Normalizing Flows	< .001	0.868	Large	Normalizing Flows > DiCE
Percentage of change	DiCE vs. Proto	.003	0.319	Medium	Proto > DiCE
Percentage of change	Human-generated vs. Normalizing Flows	< .001	0.868	Large	Normalizing Flows > Human-generated
Percentage of change	Human-generated vs. Proto	< .001	0.675	Large	Proto > Human-generated
Percentage of change	Normalizing Flows vs. Proto	< .001	0.868	Large	Normalizing Flows > Proto

Spearman Correlations Among Evaluation Metrics by Counterfactual Source.

Table 23: Spearman rank correlations among the key evaluation metrics, computed separately for each counterfactual source. Significant Holm-corrected correlations within each group are shown in bold. Correlations involving human agreement or human similarity are not defined for the human-generated group.

Group	Metric 1	Metric 2	Spearman's ρ	p_{Holm}
DiCE				
DiCE	Human agreement	Human similarity	0.400	< .001
DiCE	Human agreement	Training-set similarity	0.275	.028
DiCE	Human agreement	Percentage of change	0.283	.028
DiCE	Human similarity	Training-set similarity	0.846	< .001
DiCE	Human similarity	Percentage of change	0.014	.896
DiCE	Training-set similarity	Percentage of change	0.236	.0497
Human-generated				
Human-generated	Training-set similarity	Percentage of change	-0.484	< .001
Normalizing Flows				
Normalizing Flows	Human agreement	Human similarity	-0.113	.288
Normalizing Flows	Human agreement	Training-set similarity	-0.160	.263
Normalizing Flows	Human similarity	Training-set similarity	0.814	< .001
Proto				
Proto	Human agreement	Human similarity	0.150	.314
Proto	Human agreement	Training-set similarity	0.215	.126
Proto	Human agreement	Percentage of change	0.136	.314
Proto	Human similarity	Training-set similarity	0.523	< .001
Proto	Human similarity	Percentage of change	-0.252	.084
Proto	Training-set similarity	Percentage of change	-0.232	.110

Metric Distributions

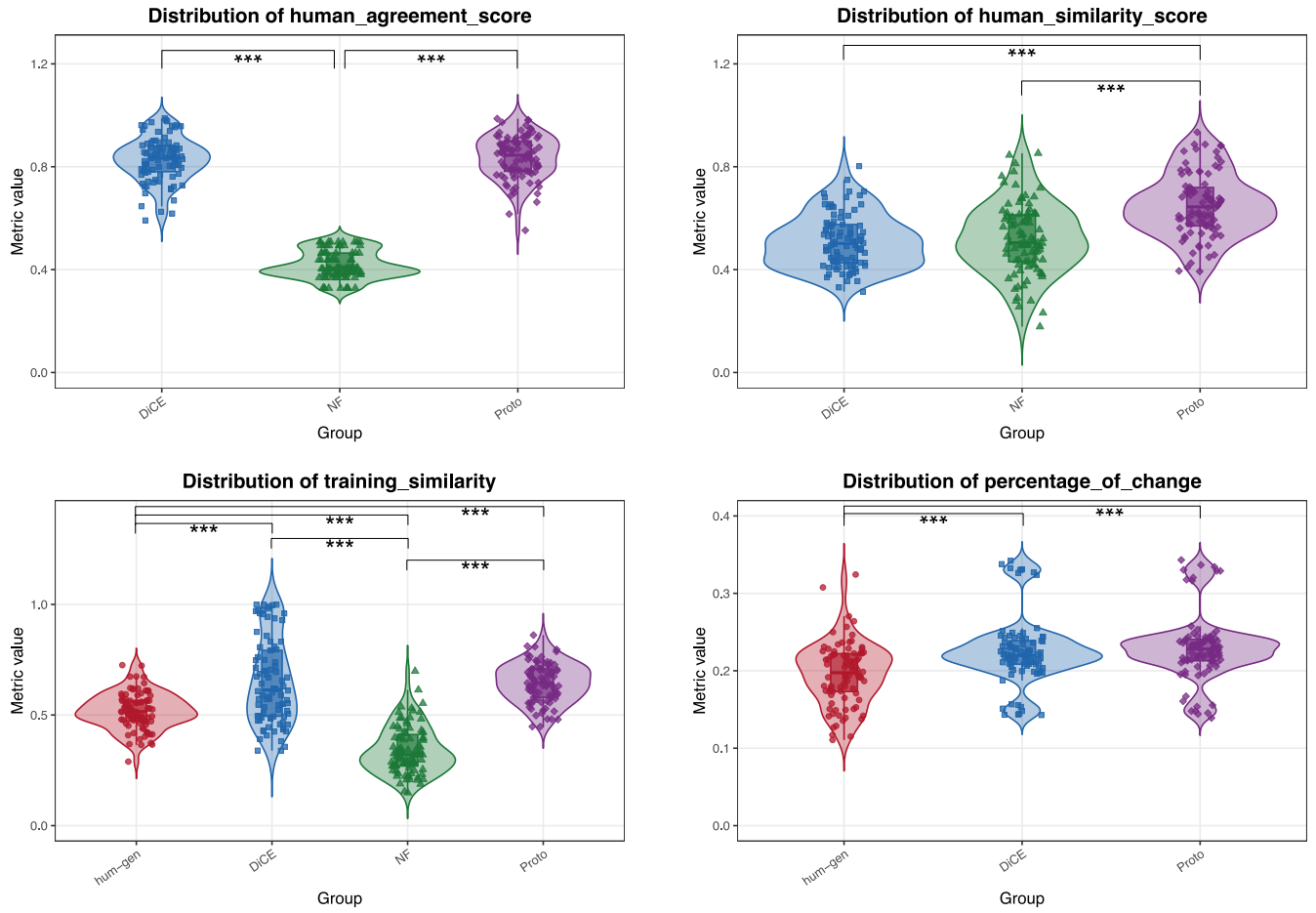


Figure 4: Distributions of the four evaluation metrics. Significance brackets denote Holm-corrected paired Wilcoxon comparisons with $*** = p_{\text{Holm}} < .001$. Top left: human agreement score. Top right: human similarity score. Bottom left: Training-set similarity, operationalized as maximum SSIM to target-class MNIST training samples. Bottom right: percentage of changed pixels. Normalized Flows was excluded from this last subplot, as percentage of change was uniform at 1.0.

Metric Relationships

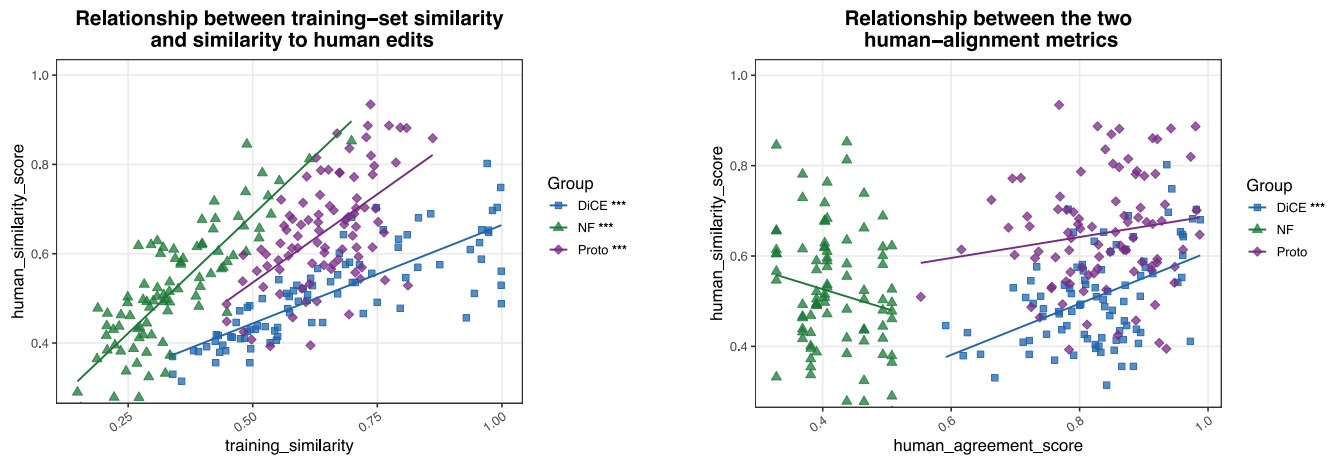


Figure 5: Key relationships among evaluation metrics. Left: Human similarity is positively associated with training-set similarity across all algorithmic sources. Right: Human similarity is associated with human agreement for DiCE only, showing that final-image similarity and change-map agreement are not interchangeable. Points indicate source–target transformations and are colored by counterfactual source. Asterisks in the legend mark significant within-source Spearman correlations after Holm correction with $*** = p_{Holm} < .001$.